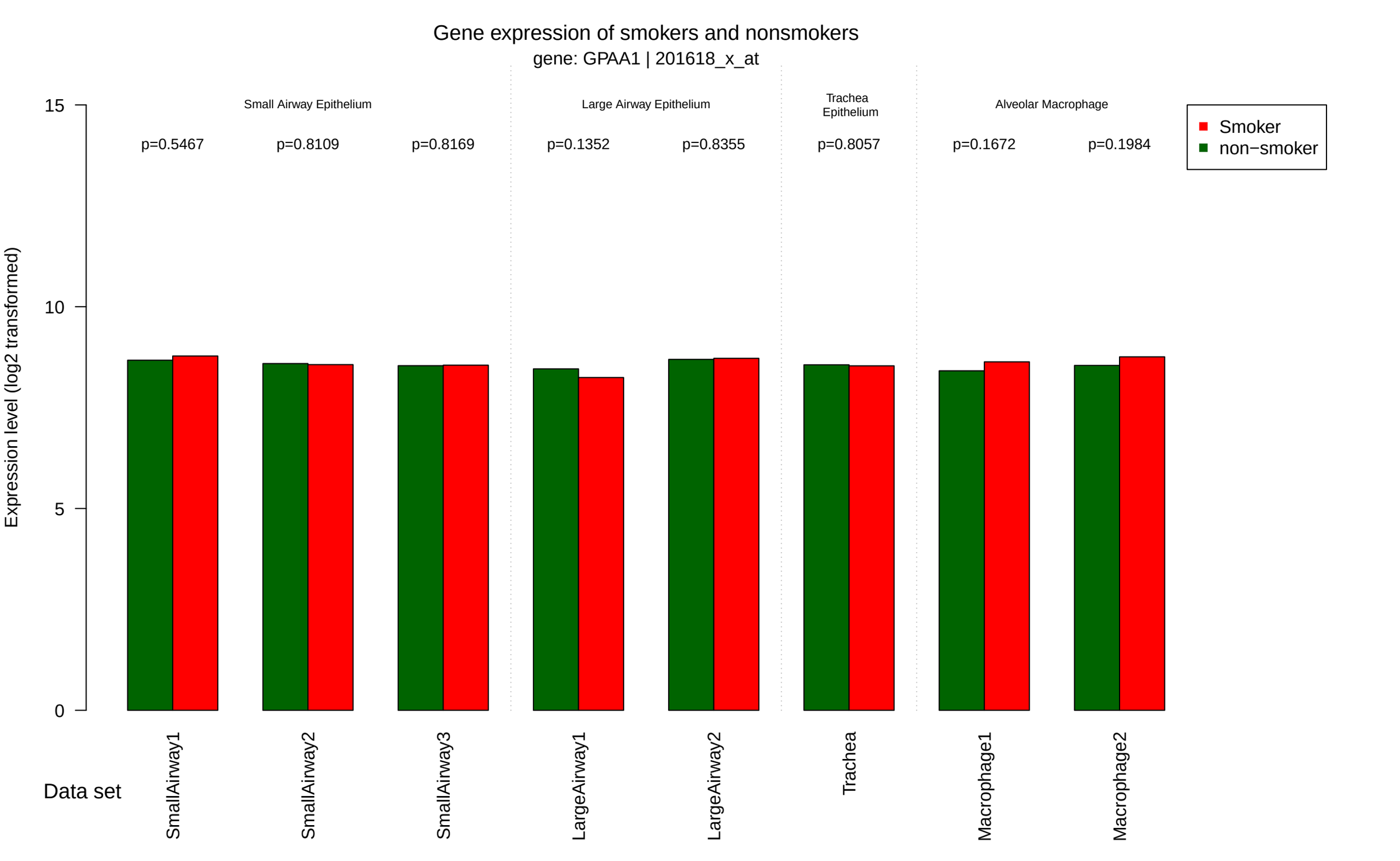
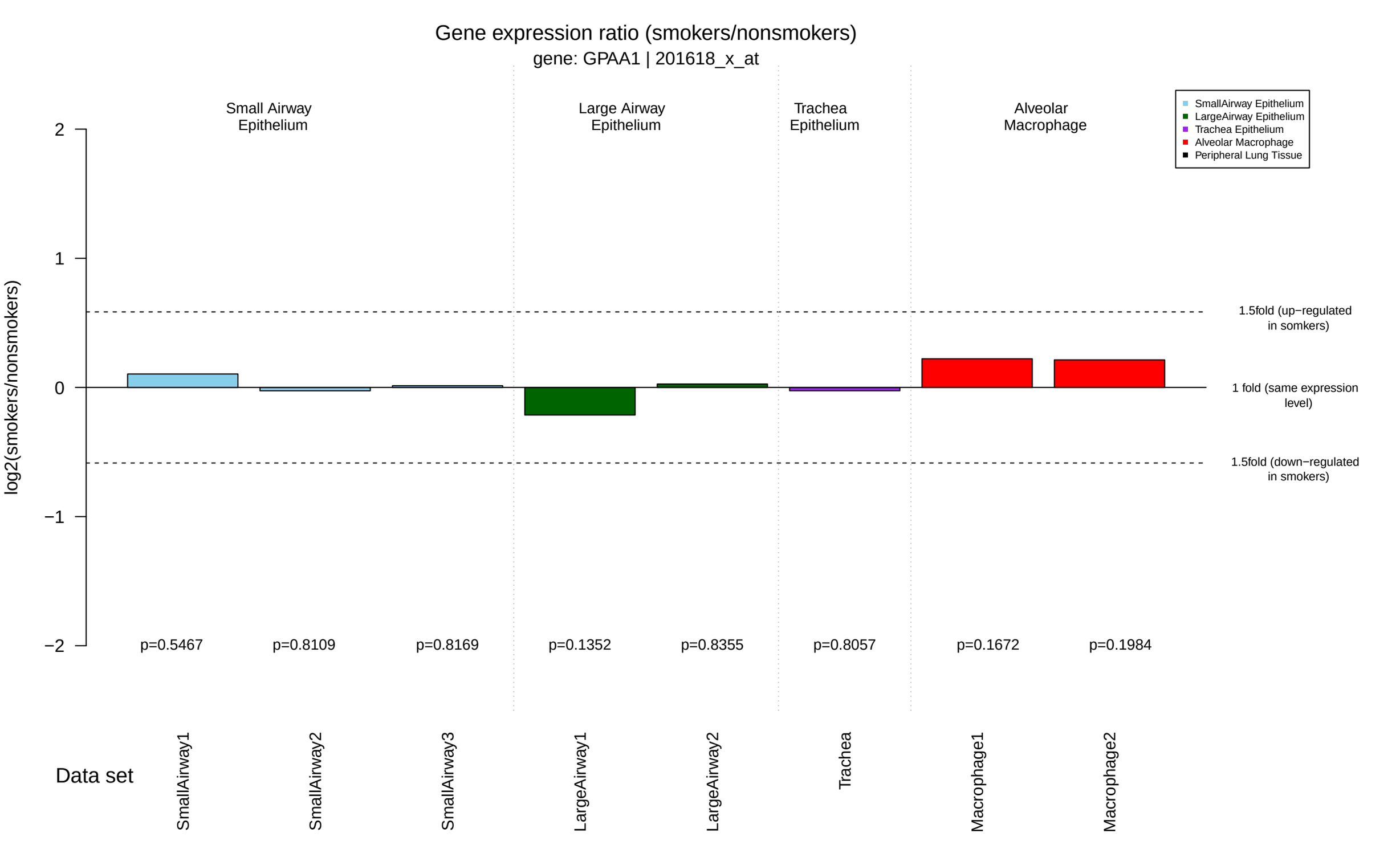
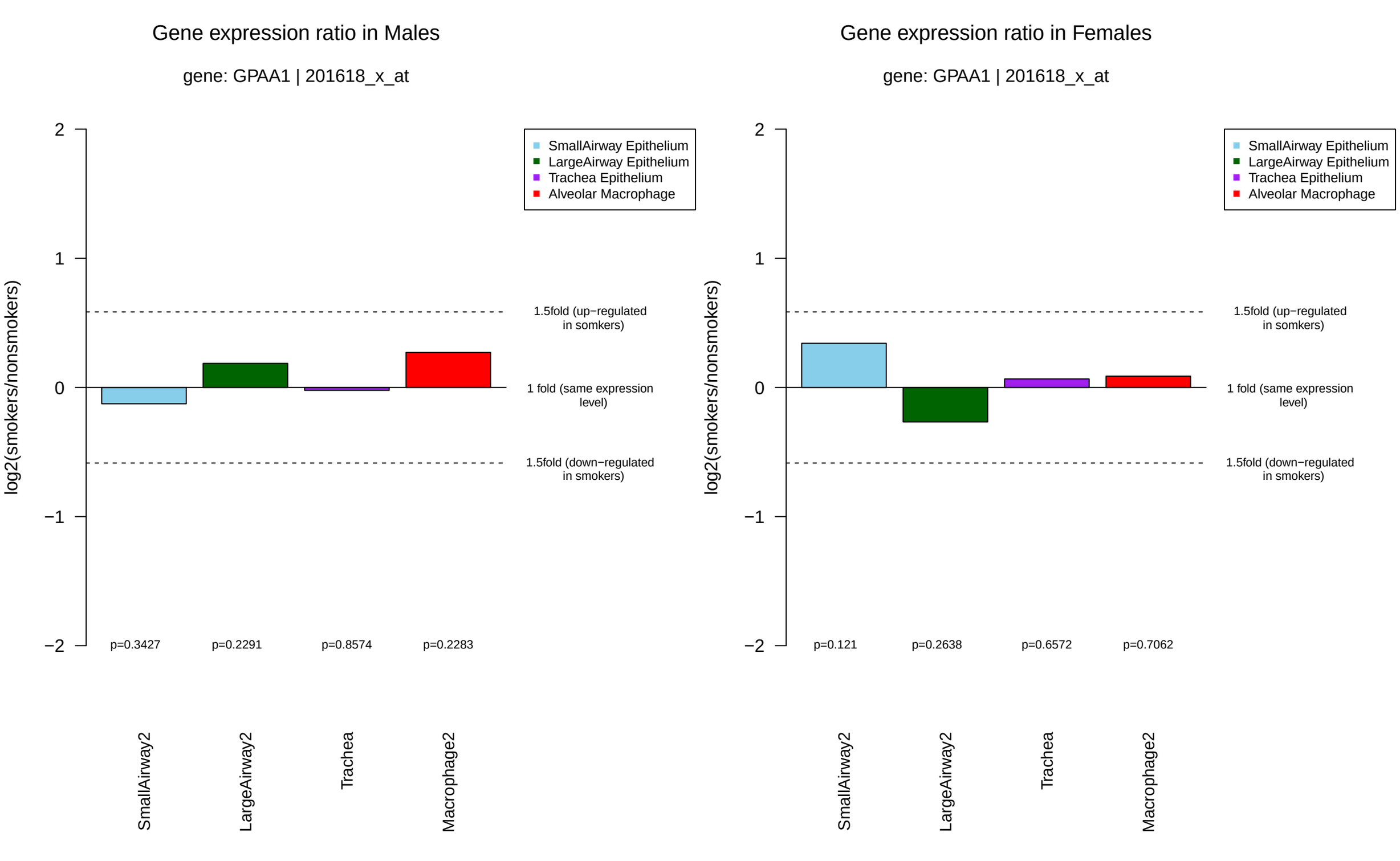
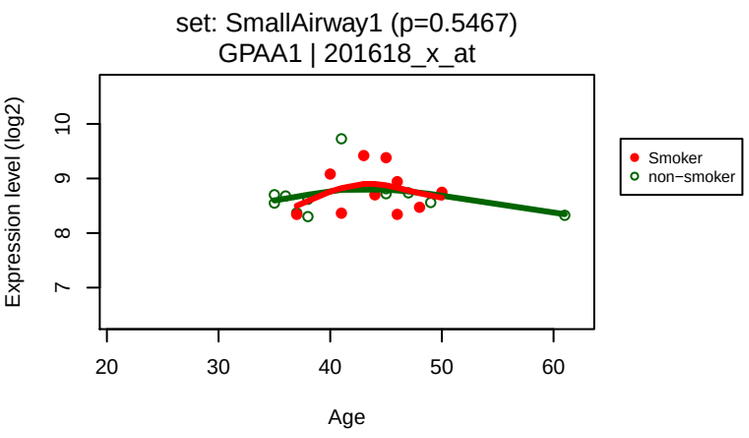
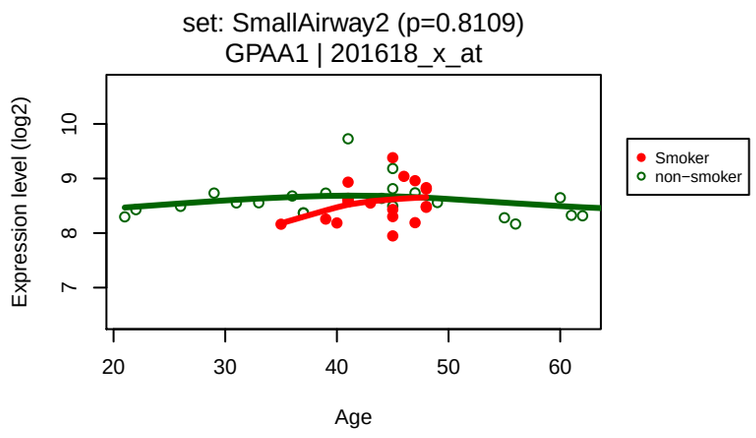
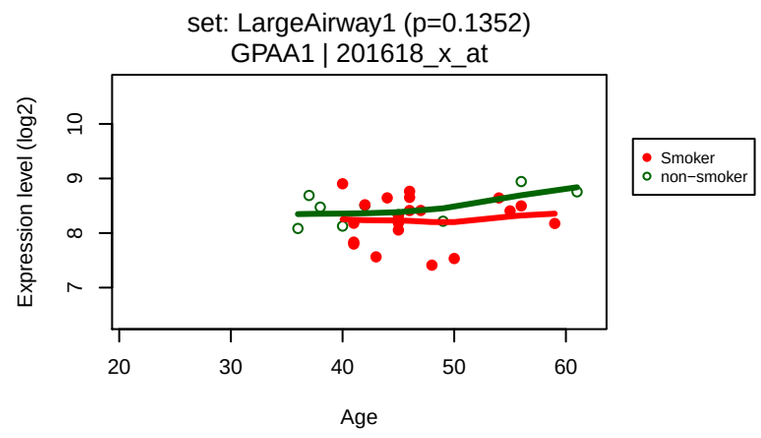

Gene expression of smokers and nonsmokers
gene: GPAA1 | 201618_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5467
p=0.8109
p=0.8169
p=0.1352
p=0.8355
p=0.8057
p=0.1672
p=0.1984
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

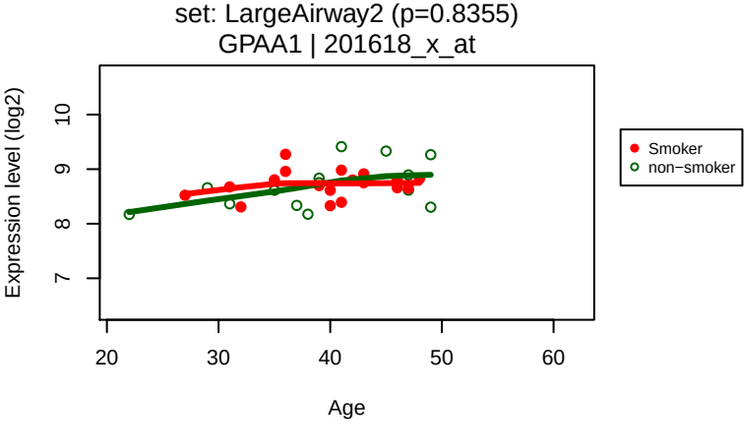

Gene expression ratio (smokers/nonsmokers)
gene: GPAA1 | 201618_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5467
p=0.8109
p=0.8169
p=0.1352
p=0.8355
p=0.8057
p=0.1672
p=0.1984
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

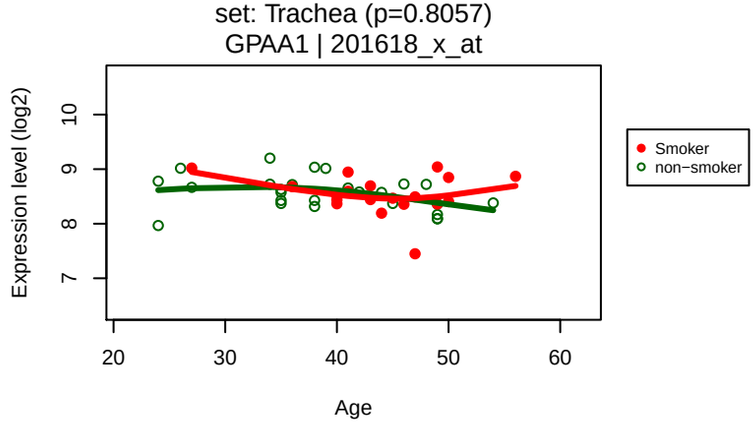

Gene expression ratio in Males
gene: GPAA1 | 201618_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3427
p=0.2291
p=0.8574
p=0.2283
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPAA1 | 201618_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.121
p=0.2638
p=0.6572
p=0.7062
SmallAirway2
LargeAirway2
Trachea
Macrophage2

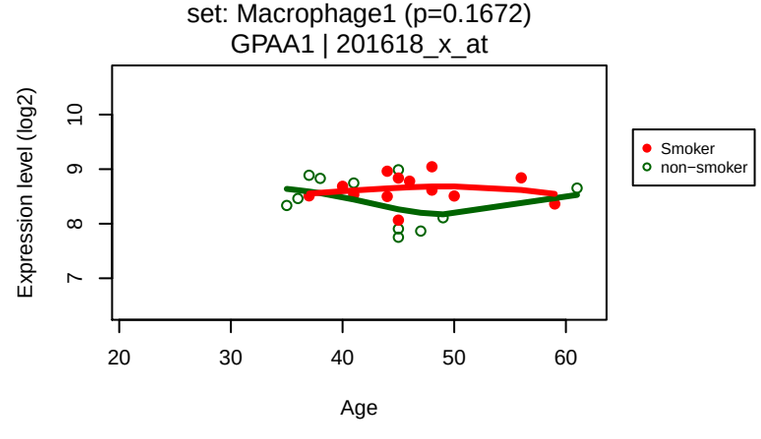

set: SmallAirway1 (p=0.5467)
GPAA1 | 201618_x_at
Expression level (log2)
Age
Smoker
non−smoker

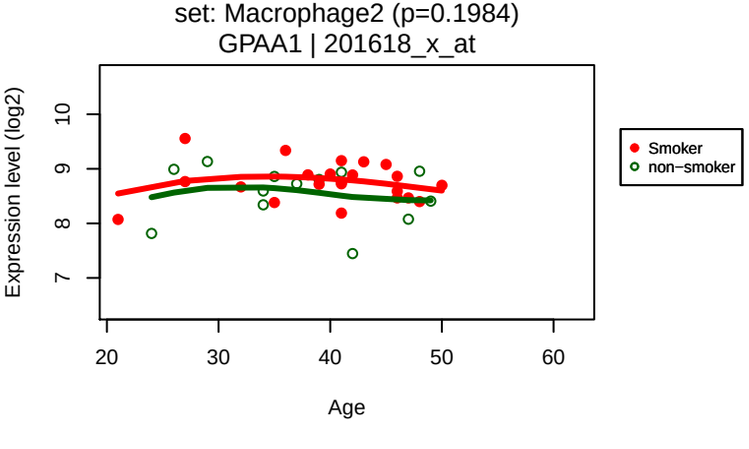

set: SmallAirway2 (p=0.8109)
GPAA1 | 201618_x_at
Expression level (log2)
Age
Smoker
non−smoker

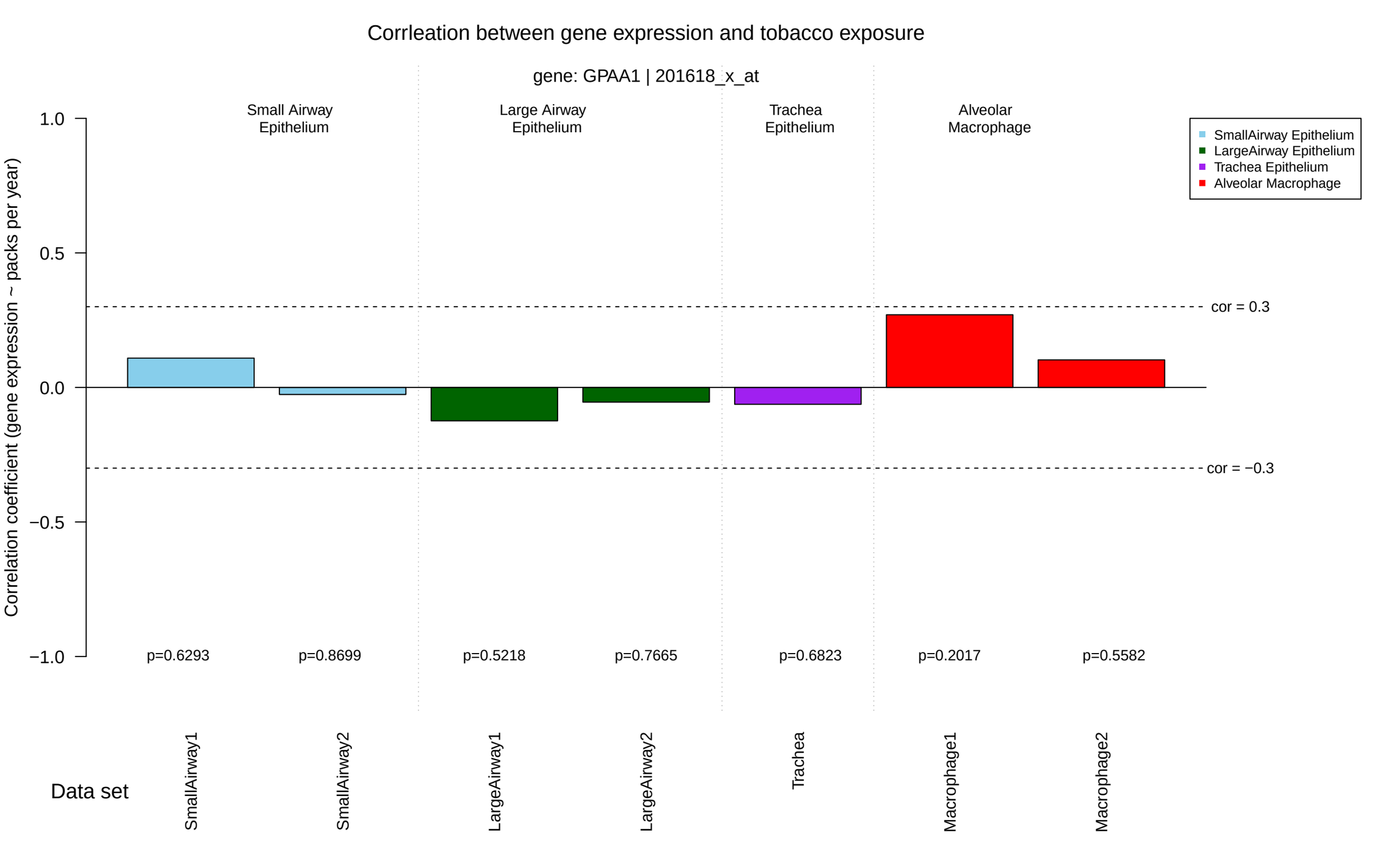

set: LargeAirway1 (p=0.1352)
GPAA1 | 201618_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8355)
GPAA1 | 201618_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8057)
GPAA1 | 201618_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1672)
GPAA1 | 201618_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1984)
GPAA1 | 201618_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPAA1 | 201618_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6293
p=0.8699
p=0.5218
p=0.7665
p=0.6823
p=0.2017
p=0.5582
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2